

A.M.N.

Aliya Maari Nilku

Project Journal

TinkerHub Useless Projects 3.0

Team: The Unplanned

Nissy – CEC

Athira – CEC

About This Journal

This journal documents the development journey of A.M.N. (Aliya Maari Nilku), an intentionally unnecessary Arduino-based personal-space enforcement device. Unlike the README, which focuses on the finished system and its technical documentation, this journal records the ideas, experiments, debugging, design decisions, abandoned features, and lessons learned while building the project.

The Idea

What problem are we solving?

We started with the idea of making something that was intentionally unnecessary. Instead of trying to solve a serious real-world problem, we asked: “What completely unnecessary problem can we over-engineer?”

The answer was personal space.

People already know how to maintain personal space. If someone stands too close, we can simply ask them to move away. But we thought: why ask them normally when we can build an electronic system to detect it and complain for us?

That became the basic idea behind A.M.N.(Aliya Maari Nilku.)

The concept was simple: detect how close someone is, determine how serious the situation is, and react dramatically.

Turning the Joke Into an Engineering Problem

Once we had the idea, we needed an actual mechanism. We decided to use an HC-SR04 ultrasonic sensor to measure distance.

Person → HC-SR04 → Arduino Uno → Distance classification → LED + Buzzer + OLED

The distance was divided into four levels:

Distance	Level	LED	Response
>25 cm	SAFE	Green	Buzzer OFF
20–25 cm	LOW	Yellow	Slow short beeps
10–20 cm	MEDIUM	Yellow	Faster beeps
≤10 cm	HIGH	Red	Rapid beeping

This gave the project an actual control system rather than just being a sensor displaying a number.

Choosing the Hardware

We chose components that were simple enough to prototype quickly while still allowing the device to have visual and audible feedback.

- Arduino Uno
- HC-SR04 ultrasonic sensor
- 0.96-inch I²C OLED display
- Green, yellow, and red LEDs
- 220 Ω current-limiting resistors
- Active buzzer
- Breadboard

- Jumper wires

The First Wiring

The major components were connected to the Arduino according to a simple pin plan.

Component	Arduino connection
Green LED	D5
Yellow LED	D6
Red LED	D7
Active buzzer	D8
HC-SR04 TRIG	D9
HC-SR04 ECHO	D10
OLED SDA	A4
OLED SCL	A5

The LEDs were connected through 220 Ω resistors, and the components shared a common ground.

Testing the OLED

The OLED was one of the components that required debugging before integration. We used an I²C scanner to determine the display's address.

Detected I²C address: 0x3C

After confirming the address, a minimal OLED test program was used. The display successfully showed test text, confirming that the OLED and its basic wiring were working.

This was an important debugging step because it separated an OLED configuration issue from the rest of the project.

Testing the LEDs

Before connecting the entire system, the LEDs were tested individually. We verified that the Arduino could control the green, yellow, and red outputs.

Testing the LEDs separately helped identify wiring and pin-assignment issues before they became mixed with the sensor and display logic.

Testing the Buzzer

We experimented with the buzzer and found that the small active buzzer could be controlled reliably using digital HIGH and LOW signals.

Instead of relying on long blocking delays, the final program uses `millis()`-based timing for the buzzer. This allows different beep patterns while the rest of the system continues to run.

- Low alert → short beep followed by a longer pause
- Medium alert → faster beeping
- High alert → rapid beeping

Making the OLED Funny

A.M.N. was supposed to be useless, so a completely serious warning system would not have been enough. We added humorous Malayalam messages to make the device feel like an unnecessarily dramatic personal-space machine.

- SAFE → “Safe aanu, so chill”
- LOW → “Aliya, aduthu aayallo”
- MEDIUM → “Machaa, konjam maaru”
- HIGH → “ALIYA MARI NILLKU!!”

The humour became an important part of the project's identity. The electronics provide the mechanism, but the exaggerated responses provide the useless-project character.

The Voice Idea That Didn't Make It

We wanted A.M.N. to be even more dramatic. Since the OLED already displayed funny Manglish warnings, we considered adding voice output so that the same warning could be spoken aloud through a laptop speaker.

Distance detected → Arduino → OLED + LED + Buzzer → USB Serial → Laptop → Text-to-Speech

The idea sounded like a perfect addition to an unnecessarily dramatic personal-space enforcement device.

However, the humour depended heavily on how the Manglish dialogue sounded. The available AI/text-to-speech pronunciation did not reproduce the intended Manglish dialogue naturally enough. The pronunciation and delivery sounded different from how we intended the dialogue to be said.

That created a problem: if the dialogue does not sound funny, the joke does not work.

We considered working around the pronunciation issue, but adding a complicated voice system just to produce a less-funny version of the original joke was not worth it. We therefore decided to drop the voice-output feature and keep the Manglish dialogue on the OLED.

This also kept the project simpler and allowed us to focus on the actual hardware prototype.

Final decision: Voice output was dropped; OLED Malayalam dialogue was retained.

This was a useful design lesson: adding more technology does not automatically make a project better. Sometimes a simpler implementation preserves the original idea more effectively.

Bringing Everything Together

Once the individual components were working, we combined them into one system.

HC-SR04 → Measure distance → Arduino Uno → OLED + LEDs + Buzzer

The system continuously measures distance and chooses an appropriate response.

The overall embedded-systems sequence is:

SENSE → PROCESS → DECIDE → ACT

1. Sense: The HC-SR04 measures the distance.
2. Process: The Arduino calculates the distance from the echo time.
3. Decide: The measured distance is compared with predefined thresholds.

4. Act: The appropriate LED, buzzer response, and OLED message are activated.

Problems We Encountered

OLED issues

At one stage, the OLED was not displaying correctly. We checked the I²C connection and used an I²C scanner. The scanner confirmed the 0x3C address. A minimal OLED program was then used to verify the display independently, after which it worked again.

Buzzer behaviour

The active buzzer worked reliably with simple digital control. We therefore kept that approach and implemented timing using `millis()` so the buzzer could produce different alert patterns without unnecessarily blocking the main loop.

Physical wiring

As more components were connected, wiring became increasingly important. LED pin assignments, resistor connections, common ground, OLED SDA/SCL, HC-SR04 TRIG/ECHO, and the buzzer connection all had to be checked carefully.

These issues reinforced the importance of testing components individually before integrating the entire circuit.

The Final Prototype

After testing the individual components and integrating them, the final prototype was assembled on a breadboard.

- Arduino Uno
- HC-SR04 ultrasonic sensor
- 0.96-inch I²C OLED
- Three status LEDs
- Active buzzer
- 220 Ω resistors
- Jumper wires

The OLED displays the detected distance and current alert level. The LEDs provide visual status and the buzzer provides an audible warning. Together, these outputs make the system progressively more dramatic as a person approaches.

Testing A.M.N.

The prototype was tested by moving an object or person progressively closer to the ultrasonic sensor.

>25 cm → SAFE
20–25 cm → LOW ALERT
10–20 cm → MEDIUM ALERT
≤10 cm → HIGH ALERT

As the distance decreased, the system escalated its visual, textual, and audible response. This was the intended behaviour.

Why Is This Useless?

At this point, we had a working embedded system. But the original question remained: does anyone actually need this?

Probably not.

If someone is standing too close, the simplest solution is: “Aliya, maari nilku.”

No ultrasonic sensor is required. No Arduino is required. No OLED is required. No buzzer is required. And definitely no three-level alert system is required.

But that is exactly what makes A.M.N. a Useless Project.

We took a simple human interaction and unnecessarily converted it into an embedded system.

What We Learned

Although the project itself is intentionally useless, building it was not.

- Arduino programming and embedded C/C++
- Digital I/O
- Ultrasonic distance measurement
- I²C communication
- OLED display control
- LED and buzzer control
- Non-blocking timing using millis()
- Breadboard prototyping
- Hardware debugging
- Hardware/software integration
- Testing and iteration
- GitHub project documentation

The project also reinforced an important engineering habit: test individual components first, then integrate them.

Final Reflection

A.M.N. started as a joke about personal space. It eventually became a complete embedded system with sensing, processing, decision-making and multiple forms of output.

The project demonstrates that even a deliberately useless idea can provide a useful opportunity to experiment with technology.

We did not make personal space safer. We did not solve an important social problem. We simply made standing too close significantly more annoying.

And that was the point.

Team Contribution

A.M.N. was developed by **The Unplanned**, with the idea, development, testing, and documentation carried out collaboratively.

Nissy - College of Engineering, Chengannur

- Initiated the A.M.N. project idea and helped shape its overall concept.
- Worked on the Arduino programming, circuit integration, and system behaviour.
- Handled the integration and testing of the ultrasonic sensor, OLED, LEDs, and buzzer.
- Worked on debugging, refinement, project documentation and demonstration.

Athira - College of Engineering, Chengannur

- Contributed to developing and refining the project concept.
- Participated in circuit assembly, component testing, and debugging.
- Helped test and refine the different alert conditions and user interaction.
- Contributed to documentation and presentation.

Working Together:

The project was developed through continuous **building, testing, debugging, and refining**. We worked together throughout the process, especially during hardware testing and integration, while dividing the programming, assembly, testing, and documentation work according to what was needed at each stage.